

Notes: Cucic and Gorea (RFS, 2026)

Nonbank Lending and the Transmission of Monetary Policy

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1 Big picture

- **Question.** When monetary policy tightens, do nonbank lenders contract like banks, or do they replace bank credit? If nonbanks do expand, how do they finance that expansion, and what does it imply for firms' and households' real outcomes?
- **Setting.** The paper studies the universe of unsecured credit to Danish firms and households from 2003 to 2018, using administrative loan-level data matched to borrower balance sheets, household tax data, and lender balance sheets.
- **Core empirical idea.** Compare how banks and nonbanks adjust credit to the same borrower after an exogenous monetary contraction. Denmark's peg to the euro lets the paper use ECB monetary policy shocks as quasi-exogenous variation for Danish rates.
- **Main result.** After an unexpected monetary tightening, nonbanks increase their credit supply relative to banks in both corporate and consumer credit markets. A one-standard-deviation shock raises the nonbank credit share by about 4% in corporate credit and about 6% in consumer credit in the preferred borrower-year fixed-effect specifications.
- **Mechanism.** The expansion is funded through long-term debt. Banks' long-term debt funding falls after a tightening, while nonbanks' long-term debt funding rises. Within the nonbank sector, lenders more reliant on long-term debt ex ante are exactly those that expand lending more.
- **Transmission to the real economy.** Nonbanks act as a "spare tire." Borrowers with pre-existing nonbank relationships suffer smaller declines in credit, investment, profits, wage bills, consumption, and asset accumulation after a monetary contraction. The benefits also spill over to other firms and households located in industries and municipalities with stronger nonbank presence.
- **What is not driving the result.** The main response operates through the intensive margin, not through a surge in new lending relationships. The paper also finds little evidence that nonbanks gain market share by taking more risk.
- **Why this paper matters.** It shows that the classic bank lending channel is no longer the full story. As nonbanks become more important, monetary contractions can be dampened by credit substitution away from banks and toward nonbanks, with uneven effects across borrowers and regions.

2 Data and measurement (key objects)

2.1 Loan-level data and sample construction

- **Source.** Mandatory annual reports to the Danish Tax Authority (SKAT) covering all unsecured credit issued during the previous 12 months.
- **Granularity.** The raw data are account-level and include borrower identity, lender identity, loan balances, and annual interest payments for products such as regular loans, credit card debt, commercial paper, and revolving credit facilities.

- **Unit of analysis.** The paper aggregates accounts to the borrower-lender-year level. For households, loans are aggregated to the household level because borrowing and consumption smoothing are household decisions.
- **Coverage.** The sample spans 2003–2018 and covers the universe of unsecured credit to Danish firms and households.
- **Main exclusions.** The paper excludes student loans, municipal loans, loans in arrears or debt forgiveness, loans issued by mortgage institutions, government bodies, extraterritorial organizations, and the Danish central bank.
- **Corporate sample restrictions.** The firm sample keeps economically active nonfinancial companies with annual sales of at least 75,000 DKK and complete accounting information, and excludes cooperatives, NGOs, and nonprofits.

2.2 Interest-rate measurement

Because contractual loan rates are not consistently reported, the paper imputes an effective annual interest rate as

$$i_{b,l,t} = \frac{\text{Interest payments}_{b,l,t}}{0.5 \times (\text{Loan balance}_{b,l,t} + \text{Loan balance}_{b,l,t-1})}. \quad (1)$$

- **Interpretation.** Interest payments are divided by the approximate average amount outstanding during the year.
- **Why it matters.** This lets the paper compare price responses across banks and nonbanks even without contractual interest-rate fields.
- **Consequence for the sample.** Because the rate measure needs the same borrower-lender pair in two consecutive years, the baseline loan-level price and quantity tests focus on pre-existing lending relationships and therefore identify intensive-margin effects.

2.3 Borrower characteristics

- **Firms.** The Danish firm register provides age, location, employment, and industry. The FIRE accounting data provide balance-sheet and income-statement variables for roughly 9,000 active businesses with more than 50 employees and a sample of smaller firms.
- **Households.** Administrative tax data provide disposable income and wealth components.
- **Consumption.** Household consumption is imputed as disposable income minus the change in net worth, following the standard Danish administrative-data approach.
- **Key firm outcomes later in the paper.** Total assets, investment, operating profits, and wage bill.
- **Key household outcomes later in the paper.** Disposable income, consumption, market value of real estate, market value of new cars, and market value of total assets.

2.4 Lender balance sheets and funding structure

- **Source.** Annual balance-sheet data from Experian for 2002–2015.
- **Objects.** Equity, short-term debt, long-term debt, and long-term funding.
- **Definition.** Long-term funding is total assets minus short-term debt.
- **Coverage.** Around 5,849 lender-year observations, of which about two-thirds are nonbanks.
- **Why this data set is crucial.** It lets the paper connect credit-supply responses at the loan level to the underlying funding model of banks and nonbanks.

2.5 Monetary policy shocks

- **Baseline shocks.** Jarocinski and Karadi (2020) high-frequency euro-area monetary policy shocks, purged of information effects.
- **Why they are usable in Denmark.** Denmark pegs its currency to the euro, so Danish monetary policy largely follows ECB decisions even when Danish conditions differ.
- **Annualization.** The paper cumulates the high-frequency shocks into an annual level measure and standardizes the resulting series.
- **Magnitude.** The annualized series ranges from about -6.4 to 17.78 basis points, with standard deviation 7.59 basis points.
- **Robustness.** The paper also uses the Altavilla et al. (2019) euro-area shock series.

2.6 Descriptive patterns that matter for the paper

- **Nonbank market share.** Nonbanks account for about 8% of unsecured credit in both corporate and consumer markets over 2003–2018.
- **Macro importance.** Total unsecured credit is roughly 50% of Danish GDP, so even a modest nonbank share matters for aggregate transmission.
- **Who the nonbanks are.** In corporate credit, specialized finance companies, wealth managers, venture-capital-style investors, and leasing companies matter most. In consumer credit, leasing companies and consumer credit companies are central.
- **Who borrows from both.** Households and firms with both bank and nonbank credit relationships are larger and more indebted than the full sample, especially in the corporate sample.
- **Funding structures differ sharply.** Banks rely heavily on short-term debt, about 70% of assets, while nonbanks rely more on equity, about 37% of assets, and long-term debt, about 30% of assets.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline loan-level specification: the shadow banking channel

The paper’s main loan-level regression is

$$y_{b,l,t} = \alpha_{b,t} + \delta_l + \beta(\text{Nonbank}_l \times \text{MP shock}_{t-1}) + \theta(\text{Nonbank}_l \times \text{Macro controls}_{t-1}) + \varepsilon_{b,l,t}. \quad (2)$$

- **Outcomes.** $y_{b,l,t}$ is either the log of outstanding debt or the effective interest rate paid by borrower b to lender l in year t .
- **Coefficient of interest.** β measures whether nonbanks change lending more than banks after a monetary shock.
- **Interpretation.** If $\beta > 0$ in the debt regression, nonbanks increase their lending share relative to banks after a monetary tightening. If $\beta > 0$ in the rate regression, nonbanks raise rates more than banks.
- **Fixed effects.** Borrower-year fixed effects control for borrower-specific demand shocks, and lender fixed effects absorb time-invariant lender characteristics.
- **What variation identifies β .** The key comparison is across lenders financing the same borrower in the same year.
- **Alternative demand control.** The paper also replaces borrower-year fixed effects with industry-location-size-time fixed effects for firms and location-income-leverage-time fixed effects for households so that single-lender borrowers enter the sample.

3.2 Funding response of lenders

To see how monetary policy affects lender funding, the paper estimates

$$\Delta \text{Funding}_{l,t} = \alpha_l + \beta \text{MP shock}_{t-1} + \theta \text{Macro controls}_{t-1} + \varepsilon_{l,t}. \quad (3)$$

- **Outcomes.** In separate regressions, Funding is equity growth, short-term debt growth, long-term debt growth, or long-term funding growth.
- **Interpretation.** A positive β means the given funding source rises after a monetary contraction.
- **Why estimated separately for banks and nonbanks.** The paper wants to show directly that the two sectors fund themselves differently after a tightening.

3.3 Which nonbanks expand? Funding structure and credit supply

The next specification links loan supply to lenders’ balance-sheet mix:

$$\begin{aligned} \log(\text{credit})_{b,l,t} = & \alpha_{b,t} + \delta_l + \beta \text{MP shock}_{t-1} + \eta \text{Funding ratio}_{l,t-1} \\ & + \theta \text{Macro controls}_{t-1} + \gamma(\text{MP shock}_{t-1} \times \text{Funding ratio}_{l,t-1}) + \varepsilon_{b,l,t}. \end{aligned} \quad (4)$$

- **Funding ratios.** The paper studies equity/TA, short-term debt/TA, long-term debt/TA, and long-term funding/TA in separate regressions.
- **Coefficient of interest.** γ tells us whether lenders that rely more on a particular funding source react more strongly to monetary shocks.
- **Prediction from the paper.** γ should be positive for long-term debt funding inside the nonbank sector.
- **Identification.** Again, the comparison is between two lenders with different funding structures lending to the same borrower in the same year.

3.4 Borrower-level financial effects

Once loans are aggregated to the borrower-year level, the paper estimates

$$\log(y_{b,t}) = \alpha_b + \beta \text{MP shock}_{t-1} + \theta \text{Macro controls}_{t-1} + \varepsilon_{b,t}. \quad (5)$$

- **Outcomes.** Total debt, total unsecured credit, bank credit, and nonbank credit.
- **Interpretation.** This specification moves from relative lending shares to absolute borrower-level credit quantities.
- **Question answered.** Does nonbank expansion fully offset the contraction in bank lending, or only partially cushion it?

3.5 Borrower-level real effects

To study whether nonbank relationships attenuate the real effects of monetary policy, the paper estimates

$$\begin{aligned} \log(y_{b,t}) = & \alpha_b + \beta(\text{Nonbank borrower}_{b,t-1} \times \text{MP shock}_{t-1}) + \gamma \text{MP shock}_{t-1} \\ & + \theta(\text{Nonbank borrower}_{b,t-1} \times \text{Macro controls}_{t-1}) + \varepsilon_{b,t}. \end{aligned} \quad (6)$$

- **Definition.** $\text{Nonbank borrower}_{b,t-1} = 1$ if more than 50% of the borrower's unsecured debt came from nonbanks in the previous year.
- **Coefficient of interest.** β measures how much pre-existing nonbank relationships soften the effect of a monetary tightening.
- **Interpretation.** A positive β means a borrower tied more strongly to nonbanks is more insulated from the contractionary shock.
- **Firm outcomes.** Total assets, investment, operating profits, and wage bill.
- **Household outcomes.** Disposable income, consumption, market value of real estate, market value of new cars, and total assets.

3.6 Risk-taking extension

In the appendix, the paper asks whether nonbanks gain share by lending to riskier borrowers:

$$y_{b,l,t} = \alpha_{b,t} + \delta_l + \beta(\text{Nonbank}_l \times \text{MP shock}_{t-1}) + \theta(\text{Nonbank}_l \times \text{Macro controls}_{t-1}) + \gamma(\text{Nonbank}_l \times \text{MP shock}_{t-1} \times \text{Borrower risk}_{b,t}) + \varepsilon_{b,l,t}. \quad (7)$$

- **Corporate risk proxies.** High leverage, high sales, and past delinquency.
- **Household risk proxies.** High debt-to-assets, disposable income, and past unemployment.
- **Prediction if there were a risk-taking channel.** γ would be positive for risky borrowers in the debt regressions.
- **Empirical result.** The paper finds no evidence that nonbanks systematically expand toward riskier firms, and in household credit the expansion is actually stronger for safer borrowers.

3.7 Conceptual model: segmented funding markets

The paper rationalizes the reduced-form evidence with a stylized model in which banks and non-banks choose lending and funding jointly. Each intermediary $i \in \{b, n\}$ solves

$$\max_{L_i, S_i, B_i} f(L_i) - c_S^i(S_i) - c_B^i(B_i) \quad \text{subject to} \quad L_i = S_i + B_i, \quad (8)$$

where L_i is lending, S_i is short-term debt, and B_i is long-term debt.

The first-order conditions imply

$$\frac{\partial f(L_i)}{\partial L_i} = \frac{\partial c_S^i(S_i)}{\partial S_i} = \frac{\partial c_B^i(B_i)}{\partial B_i}. \quad (9)$$

- **Interpretation.** In equilibrium, each lender equates the marginal return on lending with the marginal cost of short-term and long-term funding.
- **Where banks and nonbanks differ.** Banks have relatively sticky short-term deposit funding but interest-rate-sensitive long-term debt because their bond investors, such as insurers and pension funds, care about duration and capital regulation. Nonbanks have more expensive short-term wholesale funding but relatively less rate-sensitive long-term investors, such as private credit funds and asset managers.

The paper's comparative-statics expression for lending with respect to the policy rate R is

$$\frac{\partial L_i}{\partial R} = \frac{\frac{f_{LR}^i - \gamma_S^i}{c_{S,S}^i} + \frac{f_{LR}^i - \gamma_B^i}{c_{B,B}^i}}{1 - \frac{f_{LL}^i}{c_{S,S}^i} - \frac{f_{LL}^i}{c_{B,B}^i}}. \quad (10)$$

- **Notation.** f_{LR}^i is how much the marginal return on lending rises with the policy rate; γ_S^i and γ_B^i are how much the marginal costs of short- and long-term funding rise with the policy rate.
- **Economic meaning.** A tightening raises lending returns for both sectors, but it also changes the relative cost of each funding source. The sign of $\partial L_i / \partial R$ depends on which force dominates.

For nonbanks, lending rises when

$$\frac{f_{LR}^n - \gamma_B^n}{c_{B,B}^n} > \frac{\gamma_S^n - f_{LR}^n}{c_{S,S}^n}. \quad (11)$$

For banks, lending falls when

$$\frac{\gamma_B^b - f_{LR}^b}{c_{B,B}^b} > \frac{f_{LR}^b - \gamma_S^b}{c_{S,S}^b}. \quad (12)$$

- **Interpretation of the nonbank condition.** Nonbanks expand if the advantage from their relatively stable long-term funding dominates the disadvantage from more expensive short-term wholesale funding.
- **Interpretation of the bank condition.** Banks contract if the disadvantage from duration-sensitive long-term debt dominates the advantage from sticky deposits.
- **Model takeaway.** The mechanism is not that nonbanks are immune to monetary policy. It is that their funding mix makes them *less* exposed to the long-term debt margin that becomes especially painful for banks when rates rise.

4 Identification and instruments (what makes the estimates credible)

4.1 Why monetary policy is plausibly exogenous here

- **Core problem.** Monetary policy usually reacts to local macroeconomic conditions, so ordinary rate changes are endogenous to credit demand and credit supply.
- **Institutional solution.** Denmark pegs its currency to the euro and therefore shadows ECB policy rather than setting rates based on purely Danish conditions.
- **Empirical implementation.** The paper uses euro-area high-frequency monetary policy surprises as shocks to Danish monetary conditions.
- **Interpretation.** This is not a standard firm-level instrument, but the exchange-rate regime creates a plausibly exogenous source of rate variation for Denmark.

4.2 How the paper separates credit supply from credit demand

- **Borrower-year fixed effects.** In the main loan-level regressions, these absorb all time-varying borrower demand factors that are common across lenders in the same year.
- **Lender fixed effects.** These absorb time-invariant lender business-model differences.
- **Key comparison.** If the same borrower gets credit from a bank and a nonbank in the same year, differential changes after the shock are attributed to lender-side supply.

4.3 Representativeness of the multiple-lender sample

- **Potential concern.** Borrower-year fixed effects require borrowers who simultaneously borrow from both banks and nonbanks, which may not represent the full market.

- **Paper's response.** Re-estimate the main regressions with broader industry-location-size-time type fixed effects so that single-lender borrowers enter the sample.
- **Result.** The nonbank expansion remains present, though in the corporate sample the coefficients are somewhat smaller once single-lender firms are included.

4.4 Other credibility checks

- **Alternative shocks.** Results are robust to the Altavilla et al. euro-area shock series.
- **Alternative standard-error clustering and sampling.** The paper varies clustering and uses entropy balancing; the conclusions are unchanged.
- **Alternative dependent-variable transformations.** Using $\log(1 + \text{debt})$ and inverse hyperbolic sine transformations yields similar results.
- **No strong risk-taking story.** The paper shows that the nonbank response is not mainly a shift toward riskier borrowers.
- **Main limitation.** The evidence is strongest for unsecured credit, so the paper does not claim to measure the entire Danish credit system, especially mortgages.

5 Tables and figures

5.1 Figure 1: Nonbank credit share

Read:

- Nonbanks are not dominant, but they are economically meaningful throughout the sample, with average shares around 8% in both corporate and consumer credit.
- The corporate nonbank share rises after the global financial crisis, peaks around 2012–2013, then trends down. The consumer nonbank share declines until about 2010 and then climbs steadily.
- This figure matters because the paper is about a margin of substitution, not about a tiny fringe sector.

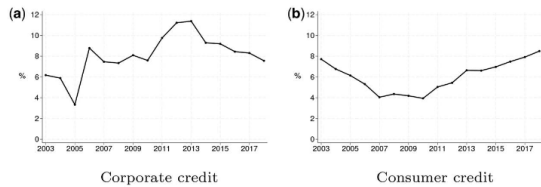


Figure 1
Nonbank credit share
Two line charts illustrate the trend of nonbank credit share from 2003 to 2018. Panel a on the left for corporate credit shows an initial rise, a peak around 2013, followed by a decline. Panel b on the right for consumer credit shows an initial decrease, then a consistent upward trend after 2010. *Source: Authors' creation based on SKAT and Statistics Denmark data.*

	Full data set				Borrowers with bank and nonbank lenders			
	(1) Count	(2) Mean	(3) Std. dev.	(4) Median	(5) Count	(6) Mean	(7) Std. dev.	(8) Median
A. Households								
Unsecured debt (thsd DKK)	30,219,386	285.25	1,802.38	64.38	7,524,172	506.58	3,012.15	201.94
Interest rate	22,440,598	0.09	0.10	0.06	7,524,172	0.10	0.07	0.08
Nonbank debt share	30,219,386	0.06	0.17	0.00	7,524,172	0.10	0.17	0.01
Number of lenders	30,219,386	2.10	1.57	2.00	7,524,172	3.71	1.52	3.00
Number of nonbank lenders	30,219,386	0.43	0.82	0.00	7,524,172	1.17	1.12	1.00
Age of oldest adult (years)	30,219,386	48.19	16.30	47.00	7,524,172	48.33	12.83	48.00
Disposable income (thsd DKK)	30,216,912	346.96	656.44	289.51	7,523,712	401.03	579.88	356.55
Recently unemployed	30,219,386	0.08	0.28	0.00	7,524,172	0.10	0.30	0.00
B. Firms								
Unsecured debt (m DKK)	1,037,047	5.22	81.30	0.31	158,841	20.08	199.75	1.70
Interest rate	842,925	0.11	0.30	0.05	158,841	0.12	0.24	0.05
Nonbank debt share	1,037,047	0.03	0.14	0.00	158,841	0.06	0.14	0.00
Number of lenders	1,037,047	1.61	0.99	1.00	158,841	2.99	1.38	3.00
Number of nonbank lenders	1,037,047	0.14	0.39	0.00	158,841	0.53	0.68	0.00
Firm age (years)	1,037,047	13.29	14.41	9.00	158,841	18.00	15.16	14.00
Debt to equity ratio	616,050	4.97	86.94	1.92	115,300	5.56	39.16	2.10
Total assets (m DKK)	719,584	68.82	1,457.57	4.01	129,536	195.95	3,214.45	11.52
FTE employees	1,037,047	22.95	351.81	2.00	158,841	96.07	855.17	7.00

This table provides descriptive statistics of borrower characteristics in the consumer and corporate credit markets between 2003–2018. Statistics are computed at the borrower-year level. Statistics in columns 1 to 4 summarize our full data set, while those in columns 5 to 8 focus on borrowers in our main estimation sample, that is, those who receive credit from both banks and nonbanks in a given year. thsd DKK stands for thousand of Danish kroner and m DKK stands for millions of Danish kroner.

5.2 Table 1: Descriptive statistics for borrowers

Read:

- Households and firms with both bank and nonbank lenders are larger, more indebted, and have more lender relationships than the average borrower.
- In the household data, dual-lender households look broadly similar in age and unemployment status to the full sample, which is why the broader-sample results are very similar to the borrower-year fixed-effect results.
- In the corporate data, dual-lender firms are much larger and older, which foreshadows why the industry-location-size-time estimates are smaller than the borrower-year fixed-effect estimates.

5.3 Table 2: Lender balance-sheet statistics

Read:

- Banks rely heavily on short-term debt, while nonbanks rely much more on equity and long-term debt.
- The funding-composition gap is central: short-term debt is about 70% of assets for banks versus 43% for nonbanks, while long-term debt is 14% for banks versus 30% for nonbanks.
- The table shows the raw balance-sheet differences that motivate the long-term debt mechanism later in the paper.

Table 2
Descriptive statistics of lenders' balance sheets

	All lenders			Nonbanks			Banks		
	Mean	Std. dev.	Median	Mean	Std. dev.	Median	Mean	Std. dev.	Median
Equity (raw growth)	0.12	0.44	0.07	0.11	0.50	0.06	0.10	0.29	0.08
Short-term debt (raw growth)	0.44	1.80	0.06	0.24	1.04	0.02	0.27	1.07	0.10
Long-term debt (raw growth)	0.19	0.80	0.02	0.14	1.04	-0.01	0.27	0.87	0.08
Lender-borrower ratio (raw growth)	0.27	0.82	0.08	0.17	1.00	0.09	-0.07	0.76	0.16
Short-term assets	0.12	0.19	0.11	0.11	0.19	0.10	0.10	0.19	0.10
Short-term debt/total assets	0.32	0.85	0.17	0.41	1.14	0.20	0.19	0.52	0.17
Long-term debt/total assets	0.22	0.72	0.14	0.20	0.81	0.12	0.16	0.51	0.12
Long-term lending/total assets	0.48	0.99	0.43	0.31	1.11	0.19	0.36	0.91	0.17

This table provides descriptive statistics of lenders. See lending variables summary (BIS and BIS). Long-term lending is reported as the difference between total assets and short-term debt. We exclude lenders with total assets below 100,000 USD and borrower growth rates at 20% per year in debt for year in June.

Table 3
Changes in nonbank lending to corporate borrowers

	(1)	(2)	(3)	(4)	(5)
A. Outcome var: Log debt					
Nonbank x MP shock	4.84*** (0.73)	4.98*** (0.55)	2.94*** (1.51)	4.09*** (0.59)	1.85** (0.94)
Observations	910,364	829,574	829,574	275,516	842,213
R2	0.19	0.80	0.80	0.65	0.40
B. Outcome var: Interest rate					
Nonbank x MP shock	-0.004*** (0.001)	-0.004*** (0.001)	-0.002*** (0.001)	-0.004** (0.002)	-0.003** (0.001)
Observations	1,119,945	1,026,918	1,026,918	380,162	782,825
R2	0.02	0.50	0.50	0.46	0.14
Macro var. interactions	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes	Yes
Lender-borrower FE		Yes	Yes		
Borrower-year FE				Yes	Yes
ILST FE					Yes

This table illustrates the results from estimating Equation (2) using as the dependent variable the log of outstanding debt (panel A) and the effective interest rate (panel B). *Nonbank* is a dummy variable equal to 1 if a lender is a financial company other than a traditional bank and equal to 0 if the lender is a traditional bank. *MP shock* are lagged monetary policy shocks from Jarocinski and Karadi (2020). *Macro var. interactions* indicates interaction terms of our nonbank lender dummy with Danish GDP, GDP forecast, and inflation as well as the VIX index for stock market uncertainty. Standard errors (in parentheses) are clustered at the borrower-lender level. Panel A reports transformed coefficients that indicate the percentage change in the nonbank lending share in response to a one-standard-deviation increase of the monetary policy measure. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 4
Changes in nonbank lending to household borrowers

	(1)	(2)	(3)	(4)	(5)
A. Outcome var: Log debt					
Nonbank x MP shock	7.26*** (0.08)	0.04 (0.05)	3.99*** (0.07)	5.77*** (0.12)	6.18*** (0.08)
Observations	29,209,158	26,260,549	26,260,549	16,171,885	28,730,149
R2	0.18	0.79	0.79	0.54	0.26
B. Outcome var: Interest rate					
Nonbank x MP shock	0.001*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.003*** (0.000)	0.002*** (0.000)
Observations	33,928,411	30,696,815	30,696,815	20,285,707	33,412,275
R2	0.10	0.59	0.59	0.50	0.12
Macro var. interactions	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes	Yes
Lender-borrower FE		Yes	Yes		
Borrower-year FE				Yes	Yes
ILST FE					Yes

This table illustrates the results from estimating Equation (2) using as the dependent variable the log of outstanding debt (panel A) and the effective interest rate (panel B). *Nonbank* is a dummy variable equal to 1 if a lender is a financial company other than a traditional bank and equal to 0 if the lender is a traditional bank. *MP shock* are lagged monetary policy shocks from Jarocinski and Karadi (2020). *Macro var. interactions* indicates interaction terms of our nonbank lender dummy with Danish GDP, GDP forecast, and inflation as well as the VIX index for stock market uncertainty. Standard errors (in parentheses) are clustered at the borrower-lender level. Panel A reports transformed coefficients that indicate the percentage change in the nonbank lending share in response to a one-standard-deviation increase of the monetary policy measure. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.4 Tables 3 and 4: Baseline changes in nonbank lending

Read:

- **Corporate credit.** In the preferred borrower-year fixed-effect specification, a one-standard-deviation monetary tightening increases the nonbank debt share by about 4.09% relative to banks.
- **Consumer credit.** The corresponding number is about 5.77% in the preferred specification, and it remains around 6.18% when broader household fixed effects are used.
- **Rates.** Price responses are small relative to quantity responses. In corporate credit, nonbanks lower rates slightly relative to banks after tightening; in consumer credit, they raise rates slightly relative to banks. The economic action is much more about quantities than prices.

5.5 Table 5: Extensive margin

Read:

- New relationships are not the main mechanism.
- In corporate credit, nonbanks are actually less likely than banks to start new relationships after a tightening.
- In consumer credit, the effect on new relationships is positive but economically tiny.
- Taken together with the baseline sample design, this says the shadow-banking channel works primarily through bigger balances to existing clients.

Table 5
Extensive margin: New lending relationships

	(1)	(2)
<i>A. Households</i>		
Nonbank x MP shock	0.003*** (0.000)	0.003*** (0.000)
Observations	36,601,369	52,187,286
R2	0.59	0.09
<i>B. Firms</i>		
Nonbank x MP shock	-0.016*** (0.002)	-0.014*** (0.001)
Observations	799,874	1,290,415
R2	0.52	0.17
Macro var. interactions	Yes	Yes
Lender FE	Yes	Yes
Borrower-year FE	Yes	
ILST FE		Yes

This table studies the extensive margin of lending through the creation of new lending relationships by estimating a linear probability model. The equation we estimate is the same as in Equation (2) of the paper. The outcome variable is an indicator variable equal to 1 if it is the first year that we observe a lending relationship between borrower b and lender l and 0 otherwise. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 6
Funding growth response to changes in monetary policy

	(1) Equity	(2) Short-term debt	(3) Long-term debt	(4) Long-term funding
<i>A. Banks</i>				
MP shock	0.03*** (0.01)	0.01 (0.02)	-0.14*** (0.02)	-0.01 (0.02)
Observations	1,517	1,514	1,044	1,514
R2	0.20	0.18	0.16	0.12
<i>B. Nonbanks</i>				
MP shock	0.04*** (0.01)	0.04 (0.05)	0.11*** (0.04)	0.05*** (0.02)
Observations	3,181	3,164	1,114	3,174
R2	0.17	0.14	0.20	0.14
Macro controls	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes

This table shows the results of estimating Equation (3) for banks and nonbanks separately. The dependent variable is the change in the funding variable indicated in the respective column title. *MP shock* are lagged monetary policy shocks from Jarocinski and Karadi (2020). *Long-term funding* is the difference between lenders' Total assets and Short-term debt. Standard errors are clustered at the lender level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.6 Table 6: Funding growth after monetary tightening

Read:

- Banks and nonbanks both show higher equity growth after a tightening, and neither shows a meaningful response in short-term debt growth.
- The decisive divergence is in long-term debt. Banks' long-term debt growth falls sharply, while nonbanks' long-term debt growth rises.
- The broader long-term funding measure tells the same story: nonbanks can replace expensive short-term funding with long-term funding, while banks cannot do so cheaply.

5.7 Table 7: Which nonbanks expand?

Read:

- This table is the strongest reduced-form mechanism result.
- In corporate credit, nonbanks with higher long-term debt-to-assets ratios expand lending more after a tightening.
- In consumer credit, long-term debt and long-term funding ratios also predict stronger lending responses, while heavier short-term debt reliance predicts weaker responses.
- This is exactly the pattern the conceptual model needs: the long-term debt-funded nonbanks are the ones that step in.

5.8 Table 8: Borrower-level financial effects

Read:

Table 7
Nonbank lending and funding structure

	(1) Equity/TA	(2) STdebt/TA	(3) LTdebt/TA	(4) LT funding/TA
<i>A. Corporate lending</i>				
MP shock x Funding ratio	-5.82 (15.58)	-7.57 (5.30)	39.05*** (17.01)	8.19 (6.20)
Observations	9,809	9,809	2,119	9,809
R2	0.82	0.82	0.73	0.82
<i>B. Consumer lending</i>				
MP shock x Funding ratio	4.66*** (0.99)	-7.65*** (0.34)	4.36*** (0.55)	8.28*** (0.39)
Observations	2,216,448	2,216,448	1,243,764	2,216,448
R2	0.63	0.63	0.63	0.63
Macro var. interactions	Yes	Yes	Yes	Yes
Lower level interactions	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes
Borrower-year FE	Yes	Yes	Yes	Yes

This table shows the results of estimating Equation (4) among nonbank lenders in corporate credit (panel A) and consumer credit (panel B) markets. The dependent variable is the log of credit at the borrower-nonbank-year level. Column names indicate the lagged variable used as *Funding ratio* in the interaction terms: "TA" refers to *Total assets*, and the abbreviations "ST" and "LT" denote, respectively, "Short-term" and "Long-term". *MP shock* are lagged monetary policy shocks from Jarociński and Karadi (2020). Standard errors are clustered at the nonbank-borrower level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 8
Financial effects of monetary policy

	(1) Debt	(2) Credit	(3) Bank Credit	(4) Nonbank Credit	(5) B. Credit Pure	(6) NB. Credit Pure
<i>A. Households</i>						
MP shock	-3.11*** (0.02)	-4.52*** (0.04)	-5.52*** (0.04)	3.94*** (0.06)	-5.53*** (0.05)	1.99*** (0.14)
Observations	22,955,365	19,719,374	18,375,312	6,385,964	12,857,536	1,076,038
R2	0.83	0.67	0.67	0.69	0.69	0.80
<i>B. Firms</i>						
MP shock	-1.66*** (0.11)	-0.05 (0.26)	-0.23 (0.27)	6.92*** (0.71)	-0.71** (0.29)	3.41*** (0.99)
Observations	663,349	692,464	666,066	77,278	606,986	20,891
R2	0.86	0.71	0.71	0.82	0.72	0.92
Macro controls	Yes	Yes	Yes	Yes	Yes	Yes
Borrower FE	Yes	Yes	Yes	Yes	Yes	Yes

This table shows the results of estimating Equation (5) for borrowing households (panel A) and firms (panel B). Data are collapsed to the borrower-year level. Outcome variables are in logs. *Debt* is computed as *Total assets* - *Equity* from the balance sheet data for firms and as total liabilities for households. *Credit* is total unsecured debt from our loan-level data. In columns 3 and 4 we separate total unsecured credit into total unsecured credit provided by banks and nonbanks, respectively. Columns 5 and 6 re-estimate columns 3 and 4 using only those borrowers who exclusively borrow from banks (B.) and nonbanks (NB.), that is, those with a nonbank debt share equal to 0 or 1. * $p < .1$; ** $p < .05$; *** $p < .01$.

- Monetary tightening reduces total debt for both households and firms, but the contraction in unsecured credit is much smaller than the contraction in total debt because nonbanks offset part of the bank pullback.
- For households, bank credit falls by about 5.5% while nonbank credit rises by about 4%.
- For firms, total unsecured credit is roughly unchanged on average because nonbank credit rises strongly even as bank credit is flat to slightly down.
- The important message is partial insulation, not full offset. Nonbanks cushion the shock but do not erase it.

5.9 Tables 9 and 10: Real effects on firms and households

Read:

- **Firms.** Monetary tightening reduces assets, investment, operating profits, and wage bills. Firms with stronger pre-existing nonbank ties are much less affected, and their investment response is actually positive on net.
- **Households.** Bank-dependent households cut disposable income, consumption, car purchases, and total assets after a tightening. Households more reliant on nonbanks are partially shielded, especially in consumption and new-car purchases.
- **Nuance.** The real-estate valuation column shows little insulation and even a slightly more negative interaction, so the main household cushioning comes from liquid spending and asset margins rather than from housing values.

Table 9
Real effects of monetary policy in corporate credit markets

	(1) Tot. Assets	(2) Investment	(3) Oper. Profit	(4) Wage Bill
MP shock	-3.66*** (0.10)	-3.95*** (0.20)	-5.84*** (0.15)	-2.50*** (0.08)
Nonbank borrower x MP shock	1.64*** (0.50)	7.30*** (1.06)	4.74*** (0.80)	0.81** (0.38)
Observations	486,830	350,364	404,948	379,772
R2	0.87	0.70	0.76	0.92
Macro control interactions	Yes	Yes	Yes	Yes
Borrower FE	Yes	Yes	Yes	Yes

This table shows the results of estimating Equation (6) for non-financial firms. Data are collapsed at the firm-year level. The dependent variables are the logarithm of total assets (column 1), investment (column 2), operating profit (column 3), and wage bill (column 4). *MP shock* represents lagged standardized monetary policy shocks from Jarociński and Karadi (2020). Nonbank borrower is a dummy equal to 1 if more than 50% of the firm's debt in the previous year came from nonbanks. All specifications include firm fixed effects and macroeconomic control interactions. Standard errors in parentheses are clustered at the borrower level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 10
Real effects of monetary policy in consumer credit markets

	(1) Disp. Income	(2) Consumption	(3) MV RE	(4) MV New Cars	(5) MV Total Assets
MP shock	-2.05*** (0.01)	-2.52*** (0.01)	-6.02*** (0.01)	-1.45*** (0.16)	-6.81*** (0.02)
Nonbank borrower x MP shock	0.23*** (0.02)	0.94*** (0.04)	-0.08** (0.04)	6.22*** (0.62)	1.21*** (0.09)
Observations	24,302,612	23,232,087	14,850,076	131,562	24,096,429
R2	0.84	0.59	0.90	0.60	0.89
Macro control interactions	Yes	Yes	Yes	Yes	Yes
Borrower FE	Yes	Yes	Yes	Yes	Yes

This table shows the results of estimating Equation (6) for households. Data are collapsed at the household-year level. The dependent variables are the logarithm of disposable income (column 1), consumption (column 2), market value of real estate (MV; column 3), market value of new cars (column 4), and the market value of total assets (column 5). *MP shock* represents standardized monetary policy shocks from Jarociński and Karadi (2020). Nonbank borrower is an indicator equal to 1 if more than 50% of the household's debt in the previous year came from nonbanks. All specifications include household fixed effects and macroeconomic control interactions. Standard errors in parentheses are clustered at the borrower level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.10 Figure E1 and Figure E2: Who the nonbanks are and where they lend

Read:

- In corporate credit, specialized finance companies are the biggest nonbank lenders; in consumer credit, leasing companies and consumer credit companies dominate.
- Nonbank corporate lending is highly concentrated in transport and storage, whereas bank lending is much more diversified across sectors.
- This matters for the later heterogeneity and spillover analysis: nonbank credit is not distributed uniformly across the economy.

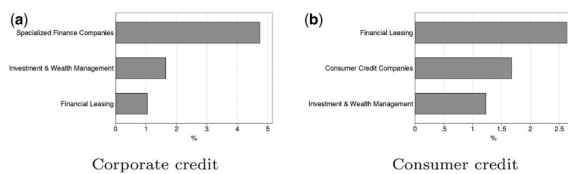


Figure E1
Largest industries among nonbank lenders
Two horizontal bar charts compare the shares of the top three nonbank lender types for corporate credit (panel a) on the left and consumer credit (panel b) on the right. For corporate credit, specialized finance companies hold the largest share, followed by investment & wealth management firms, and then financial leasing. For consumer credit, financial leasing companies hold the largest share, followed by consumer credit companies, and then investment & wealth management firms. *Source: Authors' creation based on SKAT and Statistics Denmark data.*

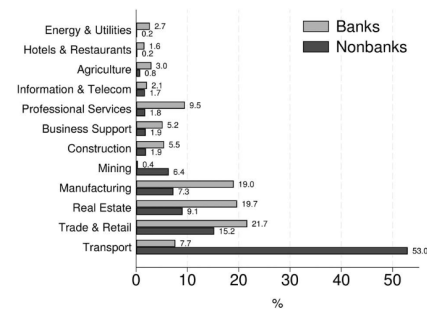


Figure E2
Distribution of banks' and nonbanks' lending across sectors
A horizontal bar chart compares the share of bank (light gray) versus nonbank (dark gray) lending across various corporate sectors. Banks concentrate heavily on the trade & retail and real estate sectors. Nonbanks, however, dominate in transport sector and have large shares in mining, manufacturing, real estate, and trade & retail. *Source: Authors' creation based on SKAT and Statistics Denmark data.*

5.11 Figure E3, Figure E4, and Table E1: Heterogeneity by place and borrower type

Read:

- Nonbank shares vary substantially across municipalities, especially in consumer credit, and the Capital/Zealand regions stand out.
- Co-borrowing from banks and nonbanks rises sharply with firm size and household income, while exclusive nonbank borrowing is more common among low-income households.
- The credit-response regressions show that medium-sized firms and higher-income households substitute more strongly toward nonbank credit after tightening.
- This implies distributional consequences: borrowers with easier access to both sectors are better able to escape the bank lending channel.

5.12 Tables E2 and E3: Heterogeneity across nonbank types

Read:

- Nonbanks are not a single homogeneous block.
- In corporate credit, the increase in nonbank lending is driven mostly by smaller “non-top-three” institutions such as venture-capital-style lenders and other credit companies.

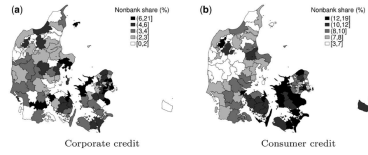


Figure E3
Nonbank share in credit supply by municipality
Two choropleth maps of Denmark illustrate the geographic distribution of nonbank credit share in corporate credit (panel a) on the left and consumer credit (panel b) on the right. For corporate credit, the highest nonbank shares are concentrated in several regions across the mainland. For consumer credit, the highest nonbank shares are concentrated in the capital region and certain areas of the mainland. Source: Authors' creation based on SKAT and Statistics Denmark data.

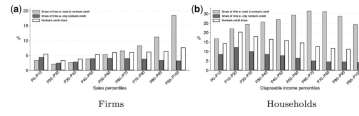


Figure E4
Heterogeneity in the borrower sample
Two grouped bar charts show borrowing patterns across firm sales deciles (panel a) on the left and household income deciles (panel b) on the right. For both firms and households, the share of borrowers that rely only on nonbank credit (dark gray bars) generally decreases as firm size or household income increases. Conversely, the share of borrowers using both bank and nonbank credit (light gray bars) generally increases with firm size or household income, indicating that larger or wealthier borrowers are more likely to use multiple credit sources. Source: Authors' creation based on SKAT and Statistics Denmark data.

- In consumer credit, leasing companies, consumer credit companies, and smaller nonbanks expand, while wealth managers do not.
- The paper argues that borrower specialization, more than big differences in funding structure across nonbank categories, explains this heterogeneity.

Table E2
Corporate lending: Nonbank heterogeneity

	(1) Fin. credit comp.	(2) Wealth managers	(3) Leasing	(4) Non top three
<i>A. Outcome var: Log debt</i>				
Nonbank x MP shock	-25.47 (17.29)	4.41 (3.38)	0.12 (2.56)	8.07*** (2.23)
Observations	186,945	201,396	212,590	232,684
R2	0.63	0.63	0.63	0.65
<i>B. Outcome var: Interest rate</i>				
Nonbank x MP shock	-0.045* (0.025)	-0.001 (0.004)	-0.001 (0.003)	-0.008*** (0.003)
Observations	260,752	280,017	298,351	319,430
R2	0.48	0.48	0.48	0.47
Macro var. interactions	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes
Borrower-year FE	Yes	Yes	Yes	Yes

This table illustrates the results from estimating Equation (2) using as the dependent variable the log of outstanding debt (panel A) and the effective interest rate (panel B). *Nonbank* is a dummy variable equal to 1 if a lender is a financial company of the category listed in column title and equal to 0 if the lender is a traditional bank. The nonbank NACE industry categories that we consider separately in each column are financial credit companies, wealth managers, leasing companies, and other nonbanks that we group in one single category called non top three. *MP shock* are lagged monetary policy shocks from Jarocinski and Karadi (2020). *Macro var. interactions* indicates interaction terms of our nonbank lender dummy with Danish GDP, GDP forecast, and inflation as well as the VIX index for stock market uncertainty. Standard errors (in parentheses) are clustered at the borrower-lender level. Panel A reports transformed coefficients that indicate the percentage change in the nonbank lending share in response to a one-standard-deviation increase of the monetary policy measure. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table E3
Consumer lending: Nonbank heterogeneity

	(1) Cons. credit comp.	(2) Wealth managers	(3) Leasing	(4) Non top three
<i>A. Outcome var: Log debt</i>				
Nonbank x MP shock	3.20*** (0.16)	-2.07*** (0.39)	8.10*** (0.20)	3.78*** (0.16)
Observations	12,244,070	8,258,883	10,920,303	12,465,345
R2	0.54	0.53	0.54	0.54
<i>B. Outcome var: Interest rate</i>				
Nonbank x MP shock	0.002*** (0.000)	0.001*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
Observations	15,643,899	10,669,539	13,751,141	15,915,491
R2	0.50	0.49	0.49	0.50
Macro var. interactions	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes
Borrower-year FE	Yes	Yes	Yes	Yes

This table illustrates the results from estimating Equation (2) using as the dependent variable the log of outstanding debt (panel A) and the effective interest rate (panel B). *Nonbank* is a dummy variable equal to 1 if a lender is a financial company of the category listed in column title and equal to 0 if the lender is a traditional bank. The nonbank NACE industry categories that we consider separately in each column are consumer credit companies, wealth managers, leasing companies, and other nonbanks that we group in one single category called non top three. *MP shock* are lagged monetary policy shocks from Jarocinski and Karadi (2020). *Macro var. interactions* indicates interaction terms of our nonbank lender dummy with Danish GDP, GDP forecast, and inflation as well as the VIX index for stock market uncertainty. Standard errors (in parentheses) are clustered at the borrower-lender level. Panel A reports transformed coefficients that indicate the percentage change in the nonbank lending share in response to a one-standard-deviation increase of the monetary policy measure. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table E1
Heterogeneous effects of monetary policy on bank and nonbank credit

	(1) In debt	(2) In credit	(3) In bank credit	(4) In nonbank credit
<i>(a) Firm-level financial outcomes</i>				
Sales p20-p40 x MP shock	-1.28 (1.72)	-0.93 (0.76)	-1.47* (0.78)	6.74*** (2.16)
Sales p40-p60 x MP shock	3.22* (1.77)	1.87*** (0.69)	1.56** (0.72)	7.25*** (1.75)
Sales p60-p80 x MP shock	3.37* (1.77)	0.48 (0.60)	0.28 (0.69)	5.78*** (1.65)
Sales p80-p100 x MP shock	1.62 (1.73)	-1.48** (0.65)	-0.63 (0.67)	4.30*** (1.57)
Observations	631,248	687,954	661,850	76,923
R2	0.86	0.72	0.71	0.82
Macro var. interactions	Yes	Yes	Yes	Yes
Borrower FE	Yes	Yes	Yes	Yes
<i>(b) Household-level financial outcomes</i>				
Disp. income p20-p40 x MP shock	0.44*** (0.09)	0.55*** (0.12)	0.60*** (0.13)	0.44** (0.18)
Disp. income p40-p60 x MP shock	-0.79*** (0.09)	0.27** (0.12)	0.32** (0.13)	1.21*** (0.18)
Disp. income p60-p80 x MP shock	-0.82*** (0.08)	0.09*** (0.11)	0.97*** (0.12)	2.11*** (0.18)
Disp. income p80-p100 x MP shock	-0.55*** (0.08)	-0.30** (0.12)	0.03 (0.13)	3.43*** (0.19)
Observations	196,776	178,967	166,609	5,715,491
R2	0.83	0.68	0.68	0.70
Macro var. interactions	Yes	Yes	Yes	Yes
Borrower FE	Yes	Yes	Yes	Yes

This table reports results of borrower-level regressions based on Equation (5), augmented with interaction terms between monetary policy shocks and indicators for borrower quantiles based on sales or disposable income. Monetary policy shocks and borrower group indicators are lagged by one period relative to the dependent variables. The dependent variables are the logarithm of total debt, unsecured credit, unsecured bank credit, and unsecured nonbank credit, respectively. Macroeconomic controls include lagged values of Danish GDP, GDP forecasts, inflation, and the VIX index. Standard errors are clustered at the borrower level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.13 Tables E5 and E6: Spillovers

Read:

- Firms in industries with larger nonbank credit shares increase investment more after a tightening, even when they themselves borrow only from banks.
- Households in municipalities with stronger nonbank presence also show better consumption outcomes after a tightening, including bank-only households once richer fixed effects are used.
- These tables are the evidence that nonbanks matter beyond their direct clients. They create positive industry and geographic spillovers.

5.14 Tables E7–E9: Risk and delinquencies

Read:

Table E5

Corporate investment and industry-level presence of nonbanks

	(1)	(2)	(3)	(4)
<i>A. All borrowers</i>				
Nonbank industry share x MP shock	5.983*** (0.318)	5.497*** (0.326)	1.038*** (0.312)	1.255*** (0.274)
Observations	658,700	658,700	658,699	613,167
R2	0.00	0.02	0.10	0.69
<i>B. Only bank borrowers</i>				
Nonbank industry share x MP shock	6.484*** (0.382)	6.391*** (0.390)	1.282*** (0.370)	1.470*** (0.330)
Observations	482,019	482,019	482,019	445,920
R2	0.00	0.02	0.10	0.69
Macro var. interactions	Yes	No	No	No
Year FE	No	Yes	Yes	Yes
Industry FE	No	No	Yes	Yes
Borrower FE	No	No	No	Yes

This table presents the results of estimating Equation (D1) in our sample of corporate borrowers. Panel A reports results based on the full sample, while panel B restricts the sample to firms without nonbank lending relationships at the time of the monetary policy shock. The outcome variable is the log of firms' investment. *Nonbank industry share* denotes the (lagged and standardized) share of nonbank credit in total (bank + nonbank) credit in the firm's industry. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table E6

Household consumption and municipality-level presence of nonbanks

	(1)	(2)	(3)	(4)
<i>A. All borrowers</i>				
Nonbank municipality share x MP shock	-0.224*** (0.018)	-0.163*** (0.018)	0.234*** (0.017)	0.152*** (0.015)
Observations	28,559,585	28,559,585	28,559,585	27,052,173
R2	0.00	0.00	0.03	0.59
<i>B. Only bank borrowers</i>				
Nonbank municipality share x MP shock	-0.075*** (0.023)	-0.320*** (0.023)	0.074*** (0.022)	0.043** (0.019)
Observations	17,563,151	17,563,151	17,563,151	16,669,278
R2	0.00	0.01	0.03	0.61
Macro var. interactions	Yes	No	No	No
Year FE	No	Yes	Yes	Yes
Municipality FE	No	No	Yes	Yes
Borrower FE	No	No	No	Yes

This table presents the results of estimating Equation (D1) in our sample of household borrowers. Panel A reports results based on the full sample, while panel B restricts the sample to households without nonbank lending relationships at the time of the monetary policy shock. The outcome variable is the log of household consumption. *Nonbank municipality share* denotes the (lagged and standardized) share of nonbank credit in total (bank + nonbank) credit in the household's municipality. * $p < .1$; ** $p < .05$; *** $p < .01$.

- The nonbank expansion is not a hidden risk-shifting story.
- Nonbanks do not lend significantly more to riskier firms relative to banks after a tightening.
- In consumer credit, the response is actually stronger for safer borrowers by leverage and income metrics.
- Loans originated by nonbanks are less likely than bank loans to become delinquent after monetary tightening.

Table E7

Risk-taking channel of monetary policy in corporate credit markets

	(1)	(2)	(3)	(4)	(5)	(6)
	ln debt	int. rate	ln debt	int. rate	ln debt	int. rate
Nonbank x JK	-0.00 (1.72)	-0.00 (0.00)	1.23 (1.65)	-0.00 (0.00)	3.68** (1.45)	-0.00** (0.00)
Triple - Leverage	-2.25 (2.59)	-0.00 (0.00)				
Triple - Sales			-3.60 (2.50)	-0.01** (0.00)		
Triple - Past delinquency					-2.12 (10.07)	-0.00 (0.01)
Observations	230,349	309,780	281,161	379,426	281,161	379,426
R2	0.66	0.46	0.65	0.46	0.65	0.46
Macro var. interactions	Yes	Yes	Yes	Yes	Yes	Yes
Lower-lvl interactions	Yes	Yes	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes	Yes	Yes
Borrower-year FE	Yes	Yes	Yes	Yes	Yes	Yes

This table illustrates the results from estimating Equation (B1) using as the dependent variable the log of outstanding debt (*ln debt*) and the effective interest rate (*int rate*). *Nonbank* is a dummy variable equal to 1 if a lender is a financial company other than a traditional bank and equal to 0 if the lender is a traditional bank. *MP shock* are lagged monetary policy shocks from Javoritski and Karadi (2020). *Macro var. interactions* indicates interaction terms of our nonbank lender dummy with Danish GDP, GDP forecast, and inflation as well as the VIX index for stock market uncertainty. *Triple* corresponds to the triple interaction terms between our monetary policy shock measure, the indicator for nonbank lenders and an indicator for whether the borrower is above the median of (i) the leverage ratio and (ii) total sales. We also consider an indicator variable for riskiness that is based on past delinquencies for each borrower. That variable takes the value of 1 if the borrower was delinquent in the previous year of observation and 0 otherwise. Lower level interactions are also included in the regression model. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table E8

Risk-taking channel of monetary policy in consumer credit markets

	(1)	(2)	(3)	(4)	(5)	(6)
	ln debt	int. rate	ln debt	int. rate	ln debt	int. rate
Nonbank x JK	5.85*** (0.17)	0.00 (0.00)	3.60*** (0.14)	0.00 (0.00)	6.17*** (0.13)	0.00*** (0.00)
Triple - Leverage	-1.47*** (0.21)	0.00 (0.00)				
Triple - Income			2.92*** (0.23)	-0.00*** (0.00)		
Triple - Unemployment					-0.27 (0.41)	-0.00*** (0.00)
Observations	14,944,449	18,689,780	16,170,775	20,284,312	16,171,885	20,285,707
R2	0.54	0.51	0.54	0.51	0.54	0.50
Macro var. interactions	Yes	Yes	Yes	Yes	Yes	Yes
Lower-lvl interactions	Yes	Yes	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes	Yes	Yes
Borrower-year FE	Yes	Yes	Yes	Yes	Yes	Yes

This table illustrates the results from estimating Equation (B1) using as the dependent variable the log of outstanding debt (*ln debt*) and the effective interest rate (*int rate*). *Nonbank* is a dummy variable equal to 1 if a lender is a financial company other than a traditional bank and equal to 0 if the lender is a traditional bank. *MP shock* are lagged monetary policy shocks from Javoritski and Karadi (2020). *Macro var. interactions* indicates interaction terms of our nonbank lender dummy with Danish GDP, GDP forecast, and inflation as well as the VIX index for stock market uncertainty. *Triple* corresponds to the triple interaction terms between our monetary policy shock measure, the indicator for nonbank lenders and an indicator for whether (i) the household is above the median of the debt-to-assets ratio distribution in a given year, (ii) the household's disposable income is above the cross-sectional median in a given year, or (iii) above the median probability of having been unemployed for at least 6 months in the last 2 years. Lower level interactions are also included in the regression model. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table E9

Future delinquencies

	(1)	(2)	(3)	(4)
	Delinq t	Delinq t+1	Delinq t	Delinq t+1
<i>A. Corporate lending</i>				
Nonbank x MP shock	-0.002*** (0.000)	-0.001 (0.000)	-0.002*** (0.000)	-0.001*** (0.000)
Observations	481,151	324,724	552,589	757,340
R2	0.50	0.51	0.13	0.15
<i>B. Consumer lending</i>				
Nonbank x MP shock	-0.000** (0.000)	-0.003*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
Observations	25,358,680	17,628,801	39,612,124	30,738,685
R2	0.27	0.60	0.26	0.26
Macro var. interactions	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes
Borrower-year FE	Yes	Yes	Yes	Yes
ILST FE			Yes	Yes

This table illustrates the results from estimating Equation (2) using as the dependent variable a dummy that takes the value of one if the firm (household) in our sample is delinquent on a given loan. Column titles indicate which delinquency dummy was used as the dependent variable, taking into account the time period of delinquency denoted by t and $t+1$. *Nonbank* is a dummy variable equal to 1 if a lender is a financial company other than a traditional bank and equal to 0 if the lender is a traditional bank. *MP shock* are lagged monetary policy shocks from Javoritski and Karadi (2020). *Macro var. interactions* indicates interaction terms of our nonbank lender dummy with Danish GDP, GDP forecast, and inflation as well as the VIX index for stock market uncertainty. Standard errors (in parentheses) are clustered at the borrower-lender level. Panel A reports the coefficients estimated in the corporate credit sample, while panel B reports those from the estimation using the consumer credit sample. * $p < .1$; ** $p < .05$; *** $p < .01$.

6 Short summary

- **Conclusion.** Monetary tightening in Denmark shifts unsecured credit away from banks and toward nonbanks, and this shift weakens the contractionary effect of policy on real activity.
- **Intuition.** Nonbanks can tap long-term investors whose funding is less sensitive to interest-rate hikes than the investors that buy bank debt.
- **Empirical lesson.** To understand monetary transmission, it is not enough to study banks alone; one must study the entire credit intermediation chain and, especially, the funding side of nonbanks.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** An unexpected monetary contraction increases the nonbank share of credit in both corporate and consumer markets.
- **Magnitude.** The relative lending response is about 4% in corporate credit and about 6% in consumer credit in the preferred specifications.
- **Credibility.** The design combines quasi-exogenous ECB shocks, borrower-year fixed effects, lender fixed effects, broader fixed-effect robustness tests, and lender balance-sheet evidence.
- **Mechanism evidence.** Monetary tightening reduces banks' long-term debt funding but increases nonbanks' long-term debt funding; within nonbanks, greater reliance on long-term debt predicts larger credit expansion.
- **Real consequences.** Borrowers tied to nonbanks are less exposed to the credit-supply side of monetary tightening, and the benefits spread to nearby firms and households through industry and geographic channels.
- **Additional discipline.** The paper rules out an important alternative interpretation by showing there is little evidence of a nonbank risk-taking channel.

7.2 Economic intuition

- Monetary policy does not hit every intermediary through the same funding margin. Banks depend on long-term investors whose demand becomes more elastic when rates move, especially because insurers and pension funds manage duration.
- Nonbanks are not insulated from higher rates, but their investor base in long-term debt is less rate-sensitive, so they can keep lending when banks pull back.
- Once borrowers already have a relationship with a nonbank, that relationship becomes valuable precisely when the banking sector is tightening credit.
- Because access to nonbanks is uneven across firms, households, and regions, the attenuation of monetary transmission is also uneven.

7.3 Takeaway

This is a clean paper on how monetary policy works in a financial system with both banks and nonbanks. The headline result is not merely that nonbanks gain market share when rates rise, but that they do so for a specific funding reason: their long-term debt is less impaired by monetary tightening than bank long-term debt. That funding asymmetry makes nonbanks a real “spare tire” for the economy and implies that future monetary transmission may become weaker and more distributionally uneven as nonbank intermediation grows.